

## MATLAB TUTORIAL #2

### discrete signals

This tutorial involves using some of the matlab functions given on the website to generate and plot discrete signals. If matlab is not currently running, then start matlab, navigate to the primary directory of the environment, and execute the `init` script in the *command* window.

1. Build the *raised cosine pulse* signal

```
>> T = 0.1;           % define the sample time (time between samples)
>> t = -4:T:4;        % define the time vector over the range [-4,4]
>> x = raisedcos(t);  % define the raised cosine function
```

2. Plot the signal and annotate the plot

```
>> figure(1);          % place the plot in figure window #1
>> plot(t,x);          % plot the signal in the figure window
>> xlabel('time (sec)'); % create a horizontal axis label
>> ylabel('amplitude'); % create a vertical axis label
>> title('raised cosine signal'); % create a title
```

The fact that this is the 1st figure window can be seen by looking at the top bar of the figure window.

Note that the plot appears somewhat smooth, even though it is composed of discrete points that are connected together with straight lines. Also note that the plot goes over the range of the time vector `[-4,4]`. Also note that the plot line color is *blue* which is the default.

3. Plot the signal and place dots at the sample points

```
>> plot(t,x,'.-');    % the string '-.' says put dots in along with lines
```

Note the time range over which this signal is greater than zero. Zoom in carefully around `t=-1` and see the exact time value where the signal becomes non-zero. You zoom into a plot by clicking on the `+` magnifying glass that is just outside the plot box in the upper right corner.

After zooming in, see if you can determine the time spacing between consecutive samples. Do you see that this is the *sample time* defined above?

To zoom in *vertically* only, place your cursor over the plot, left mouse click and hold, drag the cursor directly *up* or *down* so a stretched line appears, then let go of the mouse button when you have defined the vertical zoom you wish to apply.

To zoom in *horizontally* only, place your cursor over the plot, left mouse click and hold, drag the cursor directly *left* or *right* so a stretched line appears, then let go of the mouse button when you have defined the horizontal zoom you wish to apply.

To zoom in using a *box*, place the cursor over one corner of the box you wish to zoom to, left click and hold, drag the mouse to the other corner and let go. The plot will zoom into that box. This is useful for zooming into plots where there is very specific structure you wish to see in detail in a particular part of the plot.

To make the plot go back to its original position, click the *house* in the upper right corner of the plot box. Explore the different links that are available there to see what they do.

4. Other plot options

```
>> plot(t,x,'r');      % the string 'r' says make the line color red
>> plot(t,x,'.-r');    % make the line red, place dots
```

5. For a list of line colors and line styles, get help on the plot function by typing

```
>> help plot
```

6. Decrease the sample time so the shape of the raised cosine curve becomes more clear

```
>> T = 0.001;           % 1msec spacing between the samples
>> t = -4:T:4;
>> x = raisedcos(t);
>> plot(t,x,'.-');      % there are a lot of dots here!
```

Try to zoom in to see two sample points and see if you can identify the time spacing between those points. This might take several zoom operations because the sample points are so close together.

7. Now create a *periodic square wave*

```
>> T = 0.001;           % make the sample time really small
>> t = -10:T:10;        % define a longer time vector
>> T0 = 1;              % make the period 1 sec
>> dc = 0.2;            % make the dutycycle 20%
>> x = wave_square(t,T0,dc);
>> plot(t,x);
```

Notice that one pulse begins at  $t = 0$  and ends at  $t = 0.2$ . This constitutes a pulse that is 0.2 seconds long with a period of 1 second. Therefore, the dutycycle is  $0.2/1 = 0.2$  or 20% which is what we specified.